Scope Statement & WBS

				Date	2/11/2026
				Reporting Period	2/9/2026 – 2/13/2026
Project Sponsor	Dr. Alexander Wimmer		**Project Manager**	Cameron Nelson	
Project Start Date	1/27/2026	**Project End Date**	4/30/2026	**Document Version**	1.0 (Draft)

Part 1: Scope Statement

1. Project Objectives Summary

The KSUMB Attendance System will replace the current manual, paper-based attendance workflow for the 400+ member Kansas State University Marching Band with a unified, web-based platform. The project objectives, as established in the Project Charter, are:

- **Efficiency:** Reduce administrative overhead for KSUMB staff by approximately 30 minutes per required event by automating check-in, exception handling, and reporting.
- **Accuracy & Integrity:** Eliminate errors from manual data entry and cross-referencing between binders, spreadsheets, and paper forms.
- **Accountability:** Create a transparent, audit-friendly digital record of attendance and approved exceptions to support end-of-semester grading and minimize disputes.
- **Integration:** Consolidate data silos (FileMaker Pro, Google Sheets, paper forms) into one role-based platform with distinct views for Staff, Section Leaders, and Members.

2. In-Scope Items

The following functional areas and deliverables are within the boundaries of this project:

2.1 Core Application Features

- **Digital Attendance Capture:** GPS-verified self-check-in for members, manual check-in by section leaders, and kiosk-based check-in via student ID scan or manual login.
- **Exception Management:** Digital Pink Sheet (one-off absence) and Yellow Sheet (recurring class conflict) submission, approval, and automatic rule application.
- **Configurable Tardy Logic:** Role-based tardy thresholds with additive exception modifiers, configurable globally and per-event by staff.
- **Location Management:** Pre-defined and custom (drop-a-pin) check-in locations for staff; independent section pre-check-in locations set by section leaders.
- **Notifications:** Student-facing notifications for exception approval/denial status; section leader alerts for late Pink Sheet submissions.

2.2 Role-Based Dashboard Views

- **Staff Dashboard**: Real-time attendance overview across all sections, drill-down to section and per-student history, exception approval queue, reporting and CSV/Excel export.
- **Section Leader View**: Dynamic section roster with photos, check-in status, exception info (visibility configurable per-leader by staff), manual check-in capability.
- **Member View**: Self-check-in interface and personal attendance history. No grade information displayed.
- **Kiosk Mode**: Standalone check-in station using student ID scan or manual credentials, without SSO.

2.3 Administration & Data Management

- **Event Management**: Create, edit, and delete events with name, date, call time, location, and tardy threshold overrides.
- **Roster Management**: Bulk import from existing systems (FileMaker exports), manual editing of user information, roles, sections, and attendance records.
- **Audit Logging**: Immutable audit trail of all check-in events, status changes, and exception approvals to support accountability and dispute resolution.
- **Multi-Course Support**: Tag students by course number (100/400/800) for reporting.

2.4 Non-Functional Scope

- **Performance**: Handle 400+ simultaneous check-ins at peak time (~3:00–3:30 PM).
- **Offline Resilience**: Client-side check-in caching with automatic sync when connectivity resumes, preserving original timestamp and GPS.
- **Security**: K-State SSO/Active Directory for primary authentication; FERPA-compliant data handling; strict RBAC enforcement.
- **Mobile-First**: Responsive, lightweight web interface optimized for smartphone screens and variable outdoor connectivity.
- **Sustainability**: Hosted under K-State institutional ownership; no reliance on individual student accounts.

2.5 Project Deliverables

The project will produce the following deliverables:

#	Deliverable	Description
1	**Requirements Document (v1.0)**	Completed. Defines functional and non-functional requirements with acceptance criteria.
2	**Scope Statement & WBS**	This document. Defines project boundaries and work structure.
3	**Project Schedule (Gantt Chart)**	Timeline with milestones, dependencies, and critical path. Required for Presentation #1.
4	**High-Level & Detailed Design**	Architecture, data model, UI wireframes, and technology selection rationale.
5	**Working Prototype / MVP**	Functional application covering core check-in, exception management, and dashboards.
6	**Test Plan & Results**	Test strategy, cases, and documented results including UAT with band staff.
7	**Mid-Term Report**	Progress report for sponsor and professor. Due 3/9–3/12.
8	**Handover Package**	Source code, deployment guide, admin manual, and operational documentation for K-State IT / future band staff.
9	**Final Report & Presentation**	Comprehensive project report and client presentation (4/29–5/8).

3. Out of Scope

The following items are explicitly excluded from this project phase. Any request to include them constitutes a scope change and must go through change control.

- Integration with KSIS (university Registrar) for automatic roster or schedule importing.
- Pep-band sign-up or scheduling system.

- Functionality for other K-State music programs (Wind Symphony, Concert Band, etc.).
- Calendar view for all events.
- Automated grade calculation or grade display within the system.
- Native mobile application (iOS/Android). The solution is web-based only.
- Integration with GroupMe, Slack, or other third-party messaging platforms (beyond email for the Concern Report).

4. High-Level Acceptance Criteria

The project will be considered successfully delivered when the following conditions are met:

#	Criterion	Verified By
AC-1	Members can self-check-in via GPS within the defined geofence at a rehearsal or event.	Sponsor UAT / Field Test
AC-2	Section leaders can view their section roster with photos and check in members manually.	Sponsor UAT
AC-3	Staff can approve/deny Pink and Yellow Sheets, and approved exceptions automatically update attendance status and tardy thresholds.	Sponsor UAT / Test Cases
AC-4	Staff dashboard shows real-time attendance with drill-down to section and student level, and can export reports.	Sponsor Demo
AC-5	The system handles 400+ concurrent check-ins without failure or unacceptable delay.	Load Test
AC-6	Authentication uses K-State SSO; data access is restricted by role (FERPA-compliant).	Security Review / Test
AC-7	Handover documentation is sufficient for K-State IT or future staff to maintain the system.	Sponsor Sign-Off

5. Key Constraints & Assumptions

5.1 Constrains

- **Timeline**: Project must be completed by April 30, 2026. Final presentation window is 4/29–5/8.
- **Team Size**: Single-person project team (Cameron Nelson) with limited development hours per week alongside coursework.
- **FERPA**: All student data handling must comply with FERPA and K-State data privacy policies.
- **University IT**: SSO integration and hosting arrangements require K-State IT cooperation, which is outside the project team's direct control.
- **Connectivity**: The system must function reliably outdoors at Memorial Stadium and the practice field where cellular/Wi-Fi may be variable.
- **Budget**: Low-to-zero ongoing operational cost; no expensive commercial licenses.

5.2 Assumptions

- All band members have smartphones with GPS and modern web browsers.
- K-State IT will provide SSO integration support within a reasonable timeframe.
- FileMaker Pro export data (or equivalent) will be available for roster import development and testing.
- Dr. Wimmer and Sharyn Worcester are available for requirement validation, feedback, and UAT sessions.
- Wi-Fi at outdoor locations is sufficient for most users; offline caching addresses gaps.
- The tardy thresholds (10/20/+30 min) are initial defaults to be confirmed before deployment.

Part 2: Work Breakdown Structure

The WBS below decomposes the project into phases, deliverables, and work packages. It is structured by project phase (not chronological sequence—sequencing is handled in the project schedule). Each work package traces back to requirements from the Requirements Document.

Phase 1: Project Initiation & Planning

WBS ID	Work Package	Deliverables	Req Trace
4.1	Test plan development (unit, integration, UAT)	Test plan document	All FR & NFR
4.2	Unit & integration testing	Test results / defect log	All FR & NFR
4.3	Load / performance testing (400+ concurrent check-ins)	Load test results	NFR-01, NFR-02
4.4	Security & RBAC testing	Security test results	NFR-05–08
4.5	Field testing (outdoor GPS, connectivity)	Field test report	FR-101, NFR-02, NFR-03
4.6	User acceptance testing with band staff & section leaders	UAT sign-off / feedback log	AC-1 through AC-7
4.7	Defect resolution & regression testing	Updated defect log, clean test run	All

Phase 2: Design

WBS ID	Work Package	Deliverables	Req Trace
2.1	Technology evaluation & selection (React/Next.js, hosting, DB)	Technology selection rationale document	NFR-12, NFR-14, Tech Constraints
2.2	System architecture design (frontend, backend, DB, auth)	Architecture diagram & design document	NFR-01, NFR-05, NFR-12
2.3	Data model design (students, events, attendance, exceptions)	ERD / schema documentation	FR-107, FR-304, FR-502
2.4	UI/UX wireframes & user flow diagrams	Wireframe set for all role views	FR-401–405, NFR-09–11
2.5	SSO integration research & K-State IT coordination	SSO integration plan / findings	NFR-05, NFR-06
2.6	Geofencing & location strategy design	Location management design spec	FR-101–104

Phase 3: Development

WBS ID	Work Package	Deliverables	Req Trace
3.1	Project scaffolding & CI/CD setup	Repository, build pipeline, dev/staging environments	NFR-12
3.2	Database setup & schema implementation	Running database with seed data	2.3 → FR-107, FR-502
3.3	Authentication module (SSO + kiosk fallback)	Working login for all auth paths	NFR-05, NFR-06, FR-106
3.4	Check-in engine (GPS geofence, timestamp, data capture)	Self-check-in feature	FR-101, FR-102, FR-107, FR-108
3.5	Location management (pre-defined, custom pin, section pre-check-in)	Location configuration UI & backend	FR-102, FR-103, FR-104
3.6	Tardy threshold engine (role-based, additive modifiers, per-event override)	Tardy calculation logic & admin settings	FR-201–204
3.7	Pink Sheet workflow (submission, 24-hr warning, late flag, staff approval)	Pink Sheet feature end-to-end	FR-301–303
3.8	Yellow Sheet workflow (semester submission, staff verification, auto-apply)	Yellow Sheet feature end-to-end	FR-304–306
3.9	Section leader manual check-in	Manual check-in feature on section roster	FR-105
3.10	Staff dashboard (real-time overview, drill-down, exception queue)	Staff dashboard UI & data layer	FR-401, FR-402
3.11	Section leader view (roster, photos, status, exception info toggle)	Section leader view UI	FR-403, FR-404
3.12	Member view (self-check-in, personal history)	Member view UI	FR-405, FR-406
3.13	Kiosk mode (ID scan / manual login interface)	Kiosk check-in UI	FR-106, NFR-06
3.14	Event management (CRUD, location, threshold overrides)	Event management admin UI	FR-501
3.15	Roster import & user management	Bulk import tool & user admin UI	FR-502, FR-503, FR-504
3.16	Reporting & export (CSV/Excel, filters)	Export feature	FR-407
3.17	Daily Consolidated Concern Report (automated)	Scheduled report generation	FR-408
3.18	Notification system (exception status, late submission alerts)	Notification service	FR-301 note, new reqs
3.19	Audit logging (immutable event trail)	Audit log module	New req (audit)
3.20	Offline check-in caching & sync	PWA/service worker cache logic	FR-108, NFR-03

WBS ID	Work Package	Deliverables	Req Trace
3.21	Geofence radius configurability & GPS error handling	Configurable geofence settings	FR-102, NFR-04

Phase 4: Testing & Validation

WBS ID	Work Package	Deliverables	Req Trace
4.1	Test plan development (unit, integration, UAT)	Test plan document	All FR & NFR
4.2	Unit & integration testing	Test results / defect log	All FR & NFR
4.3	Load / performance testing (400+ concurrent check-ins)	Load test results	NFR-01, NFR-02
4.4	Security & RBAC testing	Security test results	NFR-05–08
4.5	Field testing (outdoor GPS, connectivity)	Field test report	FR-101, NFR-02, NFR-03
4.6	User acceptance testing with band staff & section leaders	UAT sign-off / feedback log	AC-1 through AC-7
4.7	Defect resolution & regression testing	Updated defect log, clean test run	All

Phase 5: Deployment & Handover

WBS ID	Work Package	Deliverables	Req Trace
5.1	Production environment setup (hosting, domain, SSL)	Running production environment	NFR-12, NFR-13
5.2	Data migration (roster import, initial configuration)	Populated production database	FR-502
5.3	Deployment & smoke testing	Deployed application, smoke test results	All
5.4	Admin & operational documentation	Admin manual, deployment guide	NFR-13
5.5	Staff & section leader training / walkthrough	Training session, user guide	NFR-13
5.6	Formal handover to K-State IT / band staff	Signed handover checklist	NFR-12, NFR-13, AC-7

Phase 6: Project Management & Course Deliverables

WBS ID	Work Package	Deliverables	Req Trace
6.1	Weekly status reports & sponsor communication	Status reports (ongoing)	MIS 677 / Charter

WBS ID	Work Package	Deliverables	Req Trace
6.2	Risk & issue tracking (ongoing)	Updated risk register	Risk Plan
6.3	In-class Presentation #2 (progress update w/ Gantt)	Slide deck	MIS 677 (Week 7: 3/3–3/5)
6.4	Mid-term report	Mid-term report document	MIS 677 (Week 8: due 3/9–3/12)
6.5	In-class Presentation #3 (progress w/ Gantt update)	Slide deck	MIS 677 (Week 10: 3/26)
6.6	Practice presentations	Practice run slides	MIS 677 (Weeks 14–15)
6.7	Final client presentation (4/29–5/8)	Final presentation materials	MIS 677 (Weeks 15–16)
6.8	Final project report & reflective report	Final report, reflective report	MIS 677 (Week 17: due 5/12)
6.9	Peer evaluation	Submitted peer eval	MIS 677 (Week 17)

WBS Summary

The WBS contains 6 phases and 52 work packages:

Phase	Work Packages	Deliverables
1. Initiation & Planning	7	7
2. Design	6	6
3. Development	21	21
4. Testing & Validation	7	7
5. Deployment & Handover	6	6
6. Project Management & Course Deliverables	9	9
TOTAL	**56**	**56**